

Week 5 Activity Log

Introduction to Node.js, npm & Express Basics

Intern Name: Aqsa Aslam

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Week Duration: July 13 – July 19, 2026

Week Overview

Week 5 was the shift from the browser to the server. Sessions introduced Node.js and the fundamentals of backend development, which is the piece that connects everything built so far to an actual database and a real, deployable application later in the program.

Objectives for the Week

- Understand how the request-response cycle works on the server side
- Get comfortable with npm-based project setup and dependency management
- Build basic REST routes in Express and understand middleware

Daily Activity Breakdown

Mon, Jul 13

Session on what Node.js actually is (a JS runtime outside the browser) and how it differs from browser JavaScript.

Tue, Jul 14

Set up Node locally, explored npm, package.json, and dependency management (dependencies vs devDependencies).

Wed, Jul 15

Learned Node's core built-in modules — fs, path, http — and wrote small scripts reading/writing files with fs.

Thu, Jul 16

Built a basic HTTP server from scratch using the http module to understand requests, responses, and status codes before using a framework.

Fri, Jul 17

Introduced to Express.js — app.get/app.post, route parameters, and middleware (express.json, custom logging middleware).

Sat, Jul 18

Practiced building a few GET/POST routes in Express that return JSON, including a simple in-memory 'notes' API for practice.

Sun, Jul 19

Reviewed REST API conventions (resource naming, HTTP verbs, status codes), tested routes with Postman, and wrote up the Week 5 log.

Detailed Summary

Coming from three weeks of frontend work, thinking in terms of routes and requests instead of clicks and DOM updates took some adjustment. Building the plain Node http server before touching Express turned out to be the right call — writing raw response headers and status codes by hand made it obvious what Express is actually doing under the hood once middleware and routing were introduced, instead of it just feeling like framework magic.

Tools & Technologies Used

- Node.js
- npm
- Express.js
- Postman (for testing routes)
- VS Code (REST Client extension for quick checks)

Skills & Concepts Learned

- Setting up and managing a Node.js project with npm and package.json
- Understanding the request-response cycle and HTTP status codes
- Writing basic Express routes and middleware
- Testing API endpoints with Postman instead of only the browser address bar

Challenges Faced & How I Resolved Them

The main friction point was middleware ordering in Express — a route wasn't receiving the parsed JSON body because `express.json()` was registered after the routes instead of before them. It's a small ordering mistake, but it produced a confusing 'undefined' error that took some searching to trace back to middleware order rather than the route logic itself.

Reflection

This week reframed how I think about a web app overall — frontend and backend aren't really separate subjects, they're two sides of the same request-response conversation. That framing made the upcoming React + backend integration feel a lot less intimidating in advance.

Goals for Next Week

- Start React and get comfortable with components, props, and state
- Keep practicing Express routes so the syntax becomes second nature before the final project
- Begin thinking ahead to how a MERN app's frontend and backend will actually talk to each other